TECHNISCHE UNIVERSITÄT
ILMENAU

Institute of Automation and Systems Engineering.

Technische Universität Ilmenau | PF 10 05 65 | 98684 Ilmenau

**Qais Yousef**

Research Assistant
Technische Universität Ilmenau
Department of Computer Science and Automation
Institute of Automation and Systems Engineering
Group of Process Optimization
Helmholtzplatz 5
98693 Ilmenau
Zusebau, Room 3010
Phone: +49 3677 69-2813
qais.yousef@tu-ilmenau.de
www.tu-ilmenau.de

Ilmenau, 30. November 2025

Letter of Recommendation for Mr. Suvarup Ghosh

To Whom It May Concern,

I am pleased to acknowledge Mr. Suvarup Ghosh for his work during his Master's studies at Technische Universität Ilmenau, conducted in collaboration with DMT GmbH & Co. KG (TÜV NORD GROUP). Based on his technical ability, scientific rigor, and professional conduct, I endorse him for doctoral studies as well as for industrial research and development roles in machine learning, geoscientific analytics, and multimodal sensing.

I supervised Mr. Ghosh's Master's thesis (30 CP), titled *"Integrating Hyperspectral Imaging and X-ray Fluorescence Data for Supervised Lithological Classification and Unsupervised Geochemical Domain Identification in Drill Cores."* His project addressed complex challenges in spectral data processing, geochemical analysis, and multimodal fusion. He successfully integrated hyperspectral imaging (HSI) and X-ray fluorescence (XRF) data into a dual-purpose analytical framework, designed to classify lithologies through supervised learning and to identify statistically coherent geochemical domains via unsupervised clustering. This design is both conceptually elegant and highly relevant for modern mineral exploration.

Mr. Ghosh also built a comprehensive XRF processing pipeline involving Compton-normalized peak analysis, percentile-based mineralization definitions, and K-means clustering with PCA initialization. His percentile-driven methodology is statistically grounded and mitigates subjectivity inherent in traditional thresholding approaches. Furthermore, he designed a cross-modal validation framework that systematically relates geochemical domains to lithological classes, ensuring consistency across modalities.

Beyond technical excellence, Mr. Ghosh is articulate, thoughtful, and highly professional. His clear writing, thorough analyses, and exemplary collaboration with our industrial partner highlight his maturity and independence. He is well-prepared to contribute to advanced research and development in machine learning, scientific data analysis, and multimodal sensing, and his future endeavors are expected to benefit from the same dedication and excellence he demonstrated during his studies.

Further information can be made available upon request.

With best regards,
Qais Yousef